Calculus I

Tutorial #3 (14–18 Oct 2024)

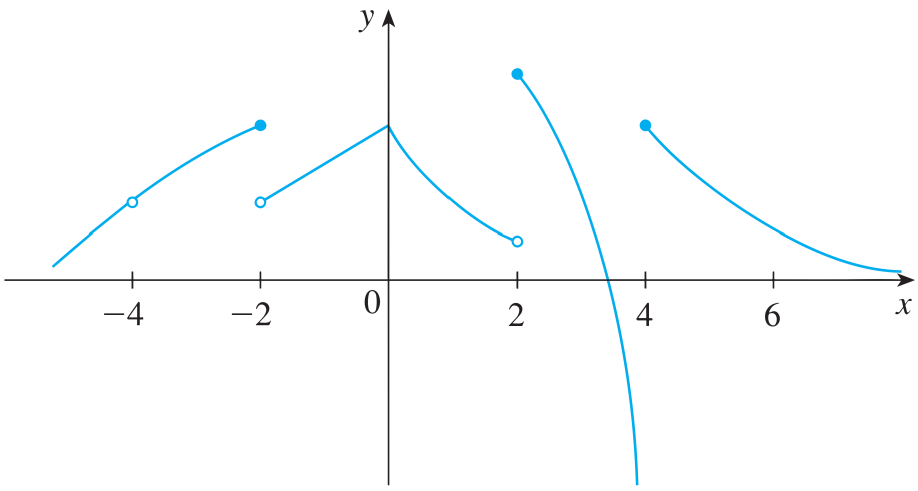
Topics:

- Continuous Functions
- Derivatives

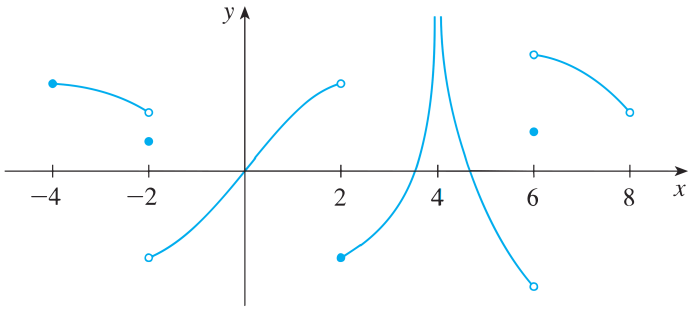
Problem 1.

a) From the graph of f, state the numbers at which f is discontinuous and explain why.

b) For each of the numbers stated in part (a), determine whether f is continuous from the right, or from the left, or neither.

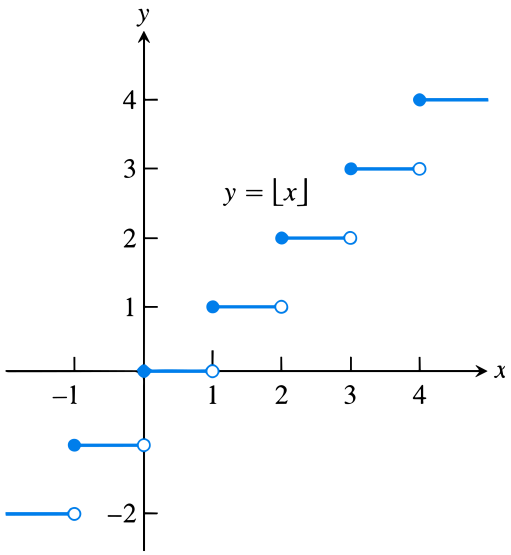


Problem 2. From the graph of g, state the intervals on which g is continuous.



Problem 3. *(The greatest integer function)* The function whose value at any number x is the greatest integer less than or equal to x is called *the greatest integer function* or *the integer floor function*. It is denoted by $\lfloor x \rfloor$ or, in some books, $[x]$.

Its graph looks as follows.



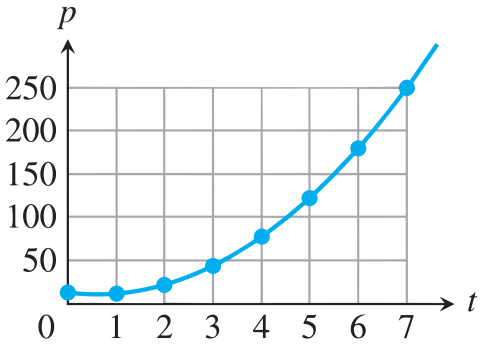
Investigate a continuity of the function $\lfloor x \rfloor$.

Problem 4. *(Removable discontinuity)* Give an example of a function $f(x)$ that is continuous for all values of x except $x = 2$, where it has a removable discontinuity. Explain how you know that f is discontinuous at $x = 2$, and how you know the discontinuity is removable.

Problem 5. Explain why the equation $x^3 - 15x + 1 = 0$ has at least three solutions in the interval $[-4, 4]$.

Problem 6. A Tibetan monk leaves the monastery at $7:00\,AM$ and takes his usual path to the top of the mountain, arriving at $7:00\,PM$. The following morning, he starts at $7:00\,AM$ at the top and takes the same path back, arriving at the monastery at $7:00\,PM$. Use the **Intermediate Value Theorem** to show that there is a point on the path that the monk will cross at exactly the same time of day on both days.

Problem 7. *(Growth of yeast cells)* In a controlled laboratory experiment, yeast cells are grown in an automated cell culture system that counts the number P of cells present at hourly intervals. The number after t hours is shown in the accompanying figure

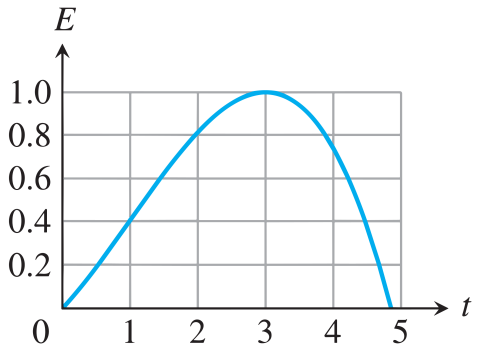


a) Explain what is meant by derivative $P'(5)$. What are its units?

b) Which is larger, $P'(2)$ or $P'(3)$? Give a reason for your answer. *Hint: Use the tangent lines.*

c) The quadratic curve capturing the trend of the data points is given by $P(t) = 6.10t^2 - 9.28t + 16.43$. Find the instantaneous rate of growth when $t = 5$ hours.

Problem 8. *(Effectiveness of a drug)* On a scale from 0 to 1, the effectiveness E of a pain-killing drug t hours after entering the blood stream is displayed in the accompanying figure.

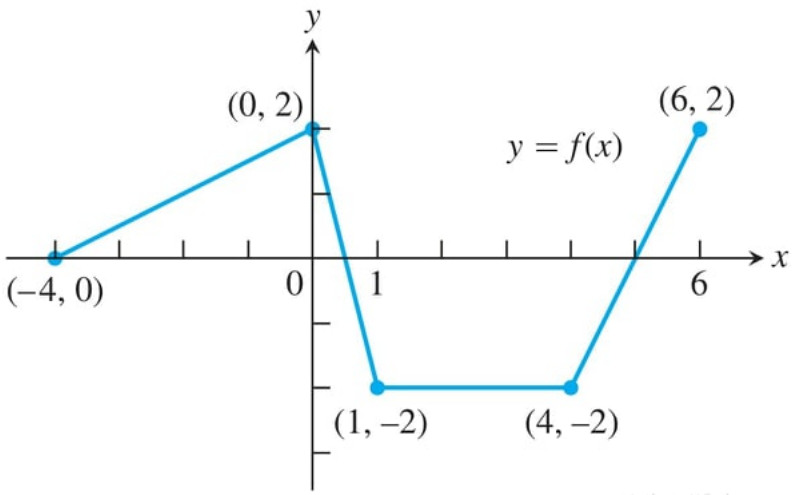


a) At what times does the effectiveness appear to be increasing? What is true about the derivative at those times?

b) At what time would you estimate that the drug reaches its maximum effectiveness? What is true about the derivative at that time? What is true about the derivative as time increases in the 1 hour *before* your estimated time?

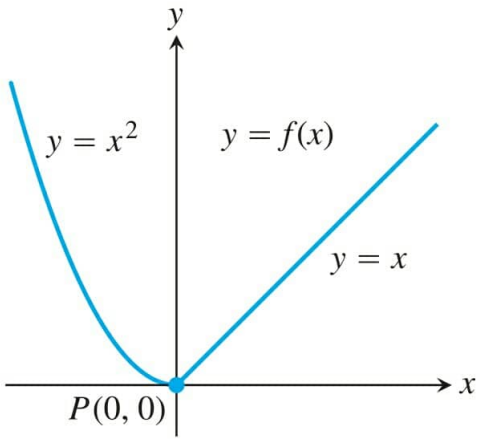
Problem 9.

a) The graph in the accompanying figure is made of line segments joined end to end. At which points of the interval [-4, 6] is f' not defined? Justify your answer.



b) Graph the derivative of f. The graph should show a step function.

Problem 10. Compare the right-hand and left-hand derivatives to show that the following function is not differentiable at the point P.



Each figure in Problems 11 and 12 shows the graph of a function over a closed interval D. At what domain points does the function appear to be

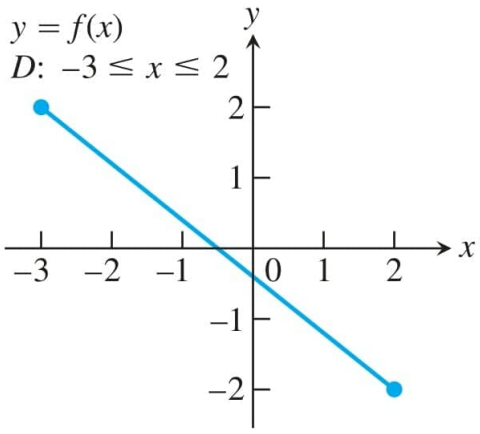
a) differentiable?

b) continuous but not differentiable?

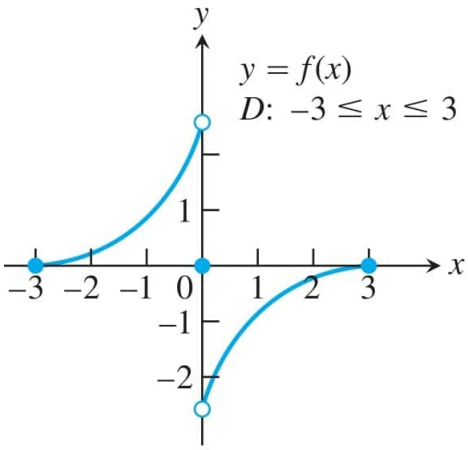
c) neither continuous nor differentiable?

Justify your answers.

Problem 11.



Problem 12.



THE END

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Answer to selected problems

1. -2, 2, and 4.

10. not differentiable at 0

12. not continuous at 0 (and then not differentiable at 0).